Jaime R. Manríquez

Applied Mathematics Ph.D.

Computational Scientist & Simulation Engineer

My Contact

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- Malmö, Sweden
- jaimemanriquezr
- Available immediately

Highlights

- 4 peer-reviewed journal papers
- 7 conference presentations, 1 invited
- University teaching since 2016, in Chile and Sweden
- 3 research grants as sole applicant

Technical Skills

- MATLAB (expert); Python, Julia (proficient); C++ (basic)
- High-order finite element methods (HDG), multiphysics coupling
- HPC clusters, batch workflows, scaling of long-running simulations
- Git, Unix shell, build systems, remote workflows, LaTeX

Languages

Spanish: Native

English: Fluent

Swedish: Conversational

Education

Ph.D. in Applied Mathematics
LTH, Lund University | 2021 – 2026

Mathematical Civil Engineering
Universidad de Concepción | 2015 – 2020
Premio Universidad de Concepción 2020, top graduating student

About Me

I am an applied mathematician with experience in numerical simulation and modelling. I enjoy working across disciplines and making technical ideas understandable to people with different backgrounds. Alongside my research, I have several years of experience teaching, supervising students and presenting scientific work. In a team, I want to take on the hard modelling problems and help colleagues get unstuck.

Experience

Researcher

Centre for Mathematical Sciences, Lund University | 2021 – 2026

- Developed a high-order finite element library in MATLAB for problems on curved geometries, including fluid–structure interaction and porous-media flow, with benchmarks in elasticity, heat transfer and cardiovascular flow
- Built a multiphysics simulator of biofilm growth and pathogen removal in drinking-water sand filters (MATLAB and Julia in 1D, Python in 2D), together with microbiologists and water engineers
- Secured three grants from the Walter Gyllenberg Foundation (2022–2025) as sole applicant, funding research visits to collaborators in Chile
- Taught exercise sessions in MATLAB and Python, and supervised bachelor's projects in mathematical modelling

Teaching Assistant

Universidad de Concepción | 2016 – 2020

- Taught numerical methods, analysis, optimisation and fluid mechanics to engineering students

Student Internships

Chile | 2019 – 2020

- **ATCAS S.A. (2020):** Predictive model of forest-fire front propagation from terrain and weather data
- **CADETECH (2019):** Non-linear optimisation of mooring-chain placement on water vessels

Publications & Engagement

Publications & Dissemination

- Peer-reviewed papers in computational mechanics and numerical analysis (CMAME, IMA J. Numer. Anal., Appl. Math. Modelling, CAMWA)
- Conference presentations, including as invited speaker
- Public presentation to high school teachers about research-level mathematics

Leadership & Academic Engagement

- Organised the department's Ph.D. seminar series; elected member of the Ph.D. student council and departmental board representative
- Undergraduate student representative at the Faculty of Engineering and the Faculty of Physical and Mathematical Sciences